ADAM: A METHOD FOR STOCHASTIC OPTIMIZATION

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ABSTRACT

We introduce *Adam*, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients. The hyper-parameters have intuitive interpretations and typically require little tuning. Some connections to related algorithms, on which *Adam* was inspired, are discussed. We also analyze the theoretical convergence properties of the algorithm and provide a regret bound on the convergence rate that is comparable to the best known results under the online convex optimization framework. Empirical results demonstrate that Adam works well in practice and compares favorably to other stochastic optimization methods. Finally, we discuss *AdaMax*, a variant of *Adam* based on the infinity norm.

1 INTRODUCTION

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems in these fields can be cast as the optimization of some scalar parameterized objective function requiring maximization or minimization with respect to its parameters. If the function is differentiable w.r.t. its parameters, gradient descent is a relatively efficient optimization method, since the computation of first-order partial derivatives w.r.t. all the parameters is of the same computational complexity as just evaluating the function. Often, objective functions are stochastic. For example, many objective functions are composed of a sum of subfunctions evaluated at different subsamples of data; in this case optimization can be made more efficient by taking gradient steps w.r.t. individual subfunctions, i.e. stochastic gradient descent (SGD) or ascent. SGD proved itself as an efficient and effective optimization method that was central in many machine learning success stories, such as recent advances in deep learning (Deng et al., 2013; Krizhevsky et al., 2012; Hinton & Salakhutdinov, 2006; Hinton et al., 2012a; Graves et al., 2013). Objectives may also have other sources of noise than data subsampling, such as dropout (Hinton et al., 2012b) regularization. For all such noisy objectives, efficient stochastic optimization techniques are required. The focus of this paper is on the optimization of stochastic objectives with high-dimensional parameters spaces. In these cases, higher-order optimization methods are ill-suited, and discussion in this paper will be restricted to first-order methods.

We propose *Adam*, a method for efficient stochastic optimization that only requires first-order gradients with little memory requirement. The method computes individual adaptive learning rates for different parameters from estimates of first and second moments of the gradients; the name *Adam* is derived from adaptive moment estimation. Our method is designed to combine the advantages of two recently popular methods: AdaGrad (Duchi et al., 2011), which works well with sparse gradients, and RMSProp (Tieleman & Hinton, 2012), which works well in on-line and non-stationary settings; important connections to these and other stochastic optimization methods are clarified in section 5. Some of Adam's advantages are that the magnitudes of parameter updates are invariant to rescaling of the gradient, its stepsizes are approximately bounded by the stepsize hyperparameter, it does not require a stationary objective, it works with sparse gradients, and it naturally performs a form of step size annealing.

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ADAM：一种用于随机优化的方法

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摘要

我们提出了*Adam*，这是一种用于随机目标函数的一阶梯度优化算法，它基于对低阶矩的自适应估计。该方法的实现十分简单，计算效率较高，内存需求较低，且对梯度的对角缩放具有不变性，因此非常适合处理数据量和/或参数量极大的问题。此外，它也适用于非平稳目标函数以及梯度存在严重噪声和/或稀疏性的问题。其超参数具有直观的含义，通常几乎不需要进行调整。文中还探讨了*Adam* 的灵感来源——一些相关算法之间的联系。同时，我们分析了该算法的理论收敛性质，并给出了一个与在线凸优化框架下现有最佳结果相当的收敛速率遗憾界。实证结果表明，Adam在实际应用中表现优异，其性能优于其他随机优化方法。最后，我们介绍了基于无穷范数的*Adam* 的变体——*AdaMax*。

1 引言

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems in these fields can be cast as the optimization of some scalar parameterized objective function requiring maximization or minimization with respect to its parameters. If the function is differentiable w.r.t. its parameters, gradient descent is a relatively efficient optimization method, since the computation of first-order partial derivatives w.r.t. all the parameters is of the same computational complexity as just evaluating the function. Often, objective functions are stochastic. For example, many objective functions are composed of a sum of subfunctions evaluated at different subsamples of data; in this case optimization can be made more efficient by taking gradient steps w.r.t. individual subfunctions, i.e. stochastic gradient descent (SGD) or ascent. SGD proved itself as an efficient and effective optimization method that was central in many machine learning success stories, such as recent advances in deep learning (Deng et al., 2013; Krizhevsky et al., 2012; Hinton & Salakhutdinov, 2006; Hinton et al., 2012a; Graves et al., 2013). Objectives may also have other sources of noise than data subsampling, such as dropout (Hinton et al., 2012b) regularization. For all such noisy objectives, efficient stochastic optimization techniques are required. The focus of this paper is on the optimization of stochastic objectives with high-dimensional parameters spaces. In these cases, higher-order optimization methods are ill-suited, and discussion in this paper will be restricted to first-order methods.

我们提出了*Adam*，这是一种高效的随机优化方法，仅需一阶梯度且内存占用极低。该方法通过梯度的一阶和二阶矩估计值，为不同参数计算出各自的自适应学习率；*Adam*这一名称正是源自“自适应矩估计”这一概念。我们的方法旨在结合两种近期较为流行的算法的优势：即擅长处理稀疏梯度的AdaGrad（Duchi等人，2011年）以及适用于在线和非平稳场景的RMSProp（Tieleman与Hinton，2012年）；第5节将对这些算法以及其他随机优化方法之间的重要关联进行详细阐述。Adam的一些优势包括：参数更新的幅度不会因梯度的重新缩放而改变，其步长大致受步长超参数的限制，无需目标函数具备平稳性，能够处理稀疏梯度，同时还天然具备某种形式的步长退火机制。

*各篇论文的贡献程度相同。作者的排序是通过在Google Hangout上抛硬币来决定的。

Algorithm 1: *Adam*, our proposed algorithm for stochastic optimization. See section 2 for details, and for a slightly more efficient (but less clear) order of computation. g_t^2 indicates the elementwise square $g_t \odot g_t$. Good default settings for the tested machine learning problems are $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. All operations on vectors are element-wise. With β_1^t and β_2^t we denote β_1 and β_2 to the power t.

Require: α: Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates for the moment estimates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0: Initial parameter vector
  $m_0 \leftarrow 0$ (Initialize 1st moment vector)
  $v_0 \leftarrow 0$ (Initialize 2nd moment vector)
  $t \leftarrow 0$ (Initialize timestep)
  while θ_t not converged **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
    $v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (Update biased second raw moment estimate)
    $\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ (Compute bias-corrected first moment estimate)
    $\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ (Compute bias-corrected second raw moment estimate)
    $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon)$ (Update parameters)
  end while
  return θ_t (Resulting parameters)

In section 2 we describe the algorithm and the properties of its update rule. Section 3 explains our initialization bias correction technique, and section 4 provides a theoretical analysis of Adam's convergence in online convex programming. Empirically, our method consistently outperforms other methods for a variety of models and datasets, as shown in section 6. Overall, we show that Adam is a versatile algorithm that scales to large-scale high-dimensional machine learning problems.

2 ALGORITHM

See algorithm 1 for pseudo-code of our proposed algorithm *Adam*. Let $f(\theta)$ be a noisy objective function: a stochastic scalar function that is differentiable w.r.t. parameters θ. We are interested in minimizing the expected value of this function, $\mathbb{E}[f(\theta)]$ w.r.t. its parameters θ. With $f_1(\theta), ..., , f_T(\theta)$ we denote the realisations of the stochastic function at subsequent timesteps $1, ..., T$. The stochasticity might come from the evaluation at random subsamples (minibatches) of datapoints, or arise from inherent function noise. With $g_t = \nabla_\theta f_t(\theta)$ we denote the gradient, i.e. the vector of partial derivatives of f_t, w.r.t θ evaluated at timestep t.

The algorithm updates exponential moving averages of the gradient (m_t) and the squared gradient (v_t) where the hyper-parameters $\beta_1, \beta_2 \in [0, 1)$ control the exponential decay rates of these moving averages. The moving averages themselves are estimates of the 1st moment (the mean) and the 2nd raw moment (the uncentered variance) of the gradient. However, these moving averages are initialized as (vectors of) 0's, leading to moment estimates that are biased towards zero, especially during the initial timesteps, and especially when the decay rates are small (i.e. the βs are close to 1). The good news is that this initialization bias can be easily counteracted, resulting in bias-corrected estimates $\widehat{m}_t$ and $\widehat{v}_t$. See section 3 for more details.

Note that the efficiency of algorithm 1 can, at the expense of clarity, be improved upon by changing the order of computation, e.g. by replacing the last three lines in the loop with the following lines: $\alpha_t = \alpha \cdot \sqrt{1 - \beta_2^t}/(1 - \beta_1^t)$ and $\theta_t \leftarrow \theta_{t-1} - \alpha_t \cdot m_t / (\sqrt{v_t} + \hat{\epsilon})$.

2.1 ADAM'S UPDATE RULE

An important property of Adam's update rule is its careful choice of stepsizes. Assuming $\epsilon = 0$, the effective step taken in parameter space at timestep t is $\Delta_t = \alpha \cdot \widehat{m}_t / \sqrt{\widehat{v}_t}$. The effective stepsize has two upper bounds: $|\Delta_t| \leq \alpha \cdot (1 - \beta_1)/\sqrt{1 - \beta_2}$ in the case $(1 - \beta_1) > \sqrt{1 - \beta_2}$, and $|\Delta_t| \leq \alpha$

算法1: *Adam*，即我们提出的用于随机优化的算法。详细内容及稍高效但略欠清晰的计算顺序请参见第2节。 g_t^2表示元素级平方运算 $g_t \odot g_t$。针对所测试的机器学习问题，良好的默认参数设置为 $\alpha = 0.001$、$\beta_1 = 0.9$、 $\beta_2 = 0.999$ 和 $\epsilon = 10^{-8}$。向量上的所有操作均为元素级操作。使用 β_1^t 和 β_2^t分别表示对 β_1 和 β_2 进行 t次幂运算。

Require: α: Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates for the moment estimates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0: Initial parameter vector
  $m_0 \leftarrow 0$ (Initialize 1st moment vector)
  $v_0 \leftarrow 0$ (Initialize 2nd moment vector)
  $t \leftarrow 0$ (Initialize timestep)
  while θ_t not converged **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
    $v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (Update biased second raw moment estimate)
    $\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ (Compute bias-corrected first moment estimate)
    $\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ (Compute bias-corrected second raw moment estimate)
    $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon)$ (Update parameters)
  end while
  return θ_t (Resulting parameters)

在第2节中，我们介绍了该算法及其更新规则的特性。第3节阐述了我们的初始化偏差校正技术，而第4节则对Adam在在线凸优化问题中的收敛性进行了理论分析。正如第6节所示，从实证角度来看，对于多种模型和数据集，我们的方法始终能展现出优于其他方法的性能。总体而言，我们证明了Adam是一种具备高度通用性的算法，能够应用于大规模高维的机器学习问题。

2算法

See algorithm 1 for pseudo-code of our proposed algorithm *Adam*. Let $f(\theta)$ be a noisy objective function: a stochastic scalar function that is differentiable w.r.t. parameters θ. We are interested in minimizing the expected value of this function, $\mathbb{E}[f(\theta)]$ w.r.t. its parameters θ. With $f_1(\theta)$ ..., , $f_T(\theta)$ we denote the realisations of the stochastic function at subsequent timesteps $1, ..., T$. The stochasticity might come from the evaluation at random subsamples (minibatches) of datapoints, or arise from inherent function noise. With $g_t = \nabla_\theta f_t(\theta)$ we denote the gradient, i.e. the vector of partial derivatives of f_t, w.r.t θ evaluated at timestep t.

该算法会更新梯度（m_t）与梯度的平方（v_t）的指数移动平均值，而超参数 $\beta_1, \beta_2 \in [0, 1$则用于控制这些移动平均值的指数衰减速率。这些移动平均值本身实际上是梯度 1st 均值（即均值）以及2nd 原始矩（即未中心化方差）的估计值。然而，由于这些移动平均值初始时被设为0向量，因此其得到的矩估计值会存在向零偏移的趋势，这种情况在初始阶段尤为明显，尤其是当衰减速率较小时（也就是 β的值接近1时）。好消息是，这种初始化偏差很容易得到纠正，从而得出经偏差校正的估计值 $\widehat{m}_t$ 和 $\widehat{v}_t$。更多细节请参见第3节。

需要注意的是，通过调整计算顺序，即用$\alpha_t = \alpha \cdot \sqrt{1 - \beta_2^t}/(1 - \beta_1^t)$和 $\theta_t \leftarrow \theta_{t-1} - \alpha_t \cdot m_t / (\sqrt{v_t} + \hat{\epsilon})$替换循环中的最后三行，可以在牺牲一定清晰度的代价下提升算法1的效率。

2.1 Adam更新规则

Adam更新规则的一个重要特性在于其对步长的精心设计。在假设 $\epsilon = 0$的前提下，时间步 t 时参数空间中实际采取的步长为 $\Delta_t = \alpha \cdot \widehat{m}_t / \sqrt{\widehat{v}_t}$。该有效步长存在两个上限：在情况$1 - \beta_1 > \sqrt{1 - \beta_2}$下为 $|\Delta_t| \leq \alpha \cdot (1 - \beta_1)/\sqrt{1 - \beta_2}$，而在其他情况下则为 $|\Delta_t| \leq \alpha$。

otherwise. The first case only happens in the most severe case of sparsity: when a gradient has been zero at all timesteps except at the current timestep. For less sparse cases, the effective stepsize will be smaller. When $(1-\beta_1) = \sqrt{1-\beta_2}$ we have that $|\widehat{m}_t/\sqrt{\widehat{v}_t}| < 1$ therefore $|\Delta_t| < \alpha$. In more common scenarios, we will have that $\widehat{m}_t/\sqrt{\widehat{v}_t} \approx \pm 1$ since $|\mathbb{E}[g]/\sqrt{\mathbb{E}[g^2]}| \leq 1$. The effective magnitude of the steps taken in parameter space at each timestep are approximately bounded by the stepsize setting α, i.e., $|\Delta_t| \lessapprox \alpha$. This can be understood as establishing a *trust region* around the current parameter value, beyond which the current gradient estimate does not provide sufficient information. This typically makes it relatively easy to know the right scale of α in advance. For many machine learning models, for instance, we often know in advance that good optima are with high probability within some set region in parameter space; it is not uncommon, for example, to have a prior distribution over the parameters. Since α sets (an upper bound of) the magnitude of steps in parameter space, we can often deduce the right order of magnitude of α such that optima can be reached from θ_0 within some number of iterations. With a slight abuse of terminology, we will call the ratio $\widehat{m}_t/\sqrt{\widehat{v}_t}$ the *signal-to-noise* ratio (SNR). With a smaller SNR the effective stepsize Δ_t will be closer to zero. This is a desirable property, since a smaller SNR means that there is greater uncertainty about whether the direction of $\widehat{m}_t$ corresponds to the direction of the true gradient. For example, the SNR value typically becomes closer to 0 towards an optimum, leading to smaller effective steps in parameter space: a form of automatic annealing. The effective stepsize Δ_t is also invariant to the scale of the gradients; rescaling the gradients g with factor c will scale $\widehat{m}_t$ with a factor c and $\widehat{v}_t$ with a factor c^2, which cancel out: $(c \cdot \widehat{m}_t)/(\sqrt{c^2 \cdot \widehat{v}_t}) = \widehat{m}_t/\sqrt{\widehat{v}_t}$.

3 INITIALIZATION BIAS CORRECTION

As explained in section 2, Adam utilizes initialization bias correction terms. We will here derive the term for the second moment estimate; the derivation for the first moment estimate is completely analogous. Let g be the gradient of the stochastic objective f, and we wish to estimate its second raw moment (uncentered variance) using an exponential moving average of the squared gradient, with decay rate β_2. Let $g_1, ..., g_T$ be the gradients at subsequent timesteps, each a draw from an underlying gradient distribution $g_t \sim p(g_t)$. Let us initialize the exponential moving average as $v_0 = 0$ (a vector of zeros). First note that the update at timestep t of the exponential moving average $v_t = \beta_2 \cdot v_{t-1} + (1-\beta_2) \cdot g_t^2$ (where g_t^2 indicates the elementwise square $g_t \odot g_t$) can be written as a function of the gradients at all previous timesteps:

$$v_t = (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} \cdot g_i^2 \tag{1}$$

We wish to know how $\mathbb{E}[v_t]$, the expected value of the exponential moving average at timestep t, relates to the true second moment $\mathbb{E}[g_t^2]$, so we can correct for the discrepancy between the two. Taking expectations of the left-hand and right-hand sides of eq. (1):

$$\mathbb{E}[v_t] = \mathbb{E}\left[(1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} \cdot g_i^2\right] \tag{2}$$

$$= \mathbb{E}[g_t^2] \cdot (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} + \zeta \tag{3}$$

$$= \mathbb{E}[g_t^2] \cdot (1-\beta_2^t) + \zeta \tag{4}$$

where $\zeta = 0$ if the true second moment $\mathbb{E}[g_t^2]$ is stationary; otherwise ζ can be kept small since the exponential decay rate β_1 can (and should) be chosen such that the exponential moving average assigns small weights to gradients too far in the past. What is left is the term $(1-\beta_2^t)$ which is caused by initializing the running average with zeros. In algorithm 1 we therefore divide by this term to correct the initialization bias.

In case of sparse gradients, for a reliable estimate of the second moment one needs to average over many gradients by chosing a small value of β_2; however it is exactly this case of small β_2 where a lack of initialisation bias correction would lead to initial steps that are much larger.

otherwise. The first case only happens in the most severe case of sparsity: when a gradient has been zero at all timesteps except at the current timestep. For less sparse cases, the effective stepsize will be smaller. When $(1-\beta_1) = \sqrt{1-\beta_2}$ we have that $|\widehat{m}_t/\sqrt{\widehat{v}_t}| < 1$ therefore $|\Delta_t| < \alpha$. In more common scenarios, we will have that $\widehat{m}_t/\sqrt{\widehat{v}_t} \approx \pm 1$ since $|\mathbb{E}[g]/\sqrt{\mathbb{E}[g^2]}| \leq 1$. The effective magnitude of the steps taken in parameter space at each timestep are approximately bounded by the stepsize setting α, i.e., $|\Delta_t| \lessapprox \alpha$. This can be understood as establishing a *trust region* around the current parameter value, beyond which the current gradient estimate does not provide sufficient information. This typically makes it relatively easy to know the right scale of α in advance. For many machine learning models, for instance, we often know in advance that good optima are with high probability within some set region in parameter space; it is not uncommon, for example, to have a prior distribution over the parameters. Since α sets (an upper bound of) the magnitude of steps in parameter space, we can often deduce the right order of magnitude of α such that optima can be reached from θ_0 within some number of iterations. With a slight abuse of terminology, we will call the ratio $\widehat{m}_t/\sqrt{\widehat{v}_t}$ the *signal-to-noise* ratio (SNR). With a smaller SNR the effective stepsize Δ_t will be closer to zero. This is a desirable property, since a smaller SNR means that there is greater uncertainty about whether the direction of $\widehat{m}_t$ corresponds to the direction of the true gradient. For example, the SNR value typically becomes closer to 0 towards an optimum, leading to smaller effective steps in parameter space: a form of automatic annealing. The effective stepsize Δ_t is also invariant to the scale of the gradients; rescaling the gradients g with factor c will scale $\widehat{m}_t$ with a factor c and $\widehat{v}_t$ with a factor c^2, which cancel out: $(c \cdot \widehat{m}_t)/(\sqrt{c^2 \cdot \widehat{v}_t}) = \widehat{m}_t/\sqrt{\widehat{v}_t}$.

3 初始化偏差校正

如第2节所述，Adam算法会运用初始化偏差校正项。此处我们将推导二阶矩估计的对应公式；一阶矩估计的推导方式与之完全类似。设 g 为随机目标函数 f的梯度，我们希望利用梯度平方的指数移动平均值，并结合衰减率 β_2，来估计该函数的第二原始矩（即未中心化方差）。令 g_1、... g_T表示后续时间步的梯度，这些梯度各自均来自底层梯度分布 $g_t \sim p(g_t)$。我们先将指数移动平均值初始化为$v_0 = 0$ （一个全零向量）。首先需要注意的是，指数移动平均值$v_t = \beta_2 \cdot v_{t-1} + (1-\beta_2) \cdot g_t^2$在时间步 t 时的更新值（其中 g_t^2表示逐元素平方 $g_t \odot g_t$ 操作）可以表示为所有先前时间步梯度值的函数：

$$v_t = (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} \cdot g_i^2 \tag{1}$$

我们希望了解在时间步 t时指数移动平均的期望值 $\mathbb{E}[v_t]$与真实的二阶矩 $\mathbb{E}[g_t^2]$之间存在怎样的关联，从而能够修正两者之间的差异。对方程(1)的左右两侧取期望值后：

$$\mathbb{E}[v_t] = \mathbb{E}\left[(1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} \cdot g_i^2\right] \tag{2}$$

$$= \mathbb{E}[g_t^2] \cdot (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} + \zeta \tag{3}$$

$$= \mathbb{E}[g_t^2] \cdot (1-\beta_2^t) + \zeta \tag{4}$$

其中，当真实的二阶矩 $\mathbb{E}[g_t^2]$ 为常数时， $\zeta = 0$ 取特定值；否则由于可以（也应当）选择适当的指数衰减率 β_1，使得指数移动平均对过于久远的梯度赋予较小的权重，因此 ζ 可以保持较小数值。此时剩下的项$(1-\beta_2^t)$则是由于用零值初始化累加平均而产生的。因此在算法1中，我们会通过除以该项来修正初始化偏差。

在梯度较为稀疏的情况下，若要获得第二矩的可靠估计，就需要通过选取较小的 β_2值对多个梯度进行平均处理；然而恰恰是在 β_2 值较小的情形下，若不进行初始化偏差校正，初始步长将会大得多。

4 CONVERGENCE ANALYSIS

We analyze the convergence of Adam using the online learning framework proposed in (Zinkevich, 2003). Given an arbitrary, unknown sequence of convex cost functions $f_1(\theta)$, $f_2(\theta)$,..., $f_T(\theta)$. At each time t, our goal is to predict the parameter θ_t and evaluate it on a previously unknown cost function f_t. Since the nature of the sequence is unknown in advance, we evaluate our algorithm using the regret, that is the sum of all the previous difference between the online prediction $f_t(\theta_t)$ and the best fixed point parameter $f_t(\theta^*)$ from a feasible set $\mathcal{X}$ for all the previous steps. Concretely, the regret is defined as:

$$R(T) = \sum_{t=1}^{T} [f_t(\theta_t) - f_t(\theta^*)] \tag{5}$$

where $\theta^* = \arg\min_{\theta \in \mathcal{X}} \sum_{t=1}^{T} f_t(\theta)$. We show Adam has $O(\sqrt{T})$ regret bound and a proof is given in the appendix. Our result is comparable to the best known bound for this general convex online learning problem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the ith element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the ith dimension of the gradients over all iterations till t, $g_{1:t,i} = [g_{1,i}, g_{2,i}, \cdots, g_{t,i}]$. Also, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ, that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 4.1. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in R^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, ..., T\}$, and β_1, $\beta_2 \in [0, 1)$ satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. Let $\alpha_t = \frac{\alpha}{\sqrt{t}}$ and $\beta_{1,t} = \beta_1 \lambda^{t-1}, \lambda \in (0, 1)$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Our Theorem 4.1 implies when the data features are sparse and bounded gradients, the summation term can be much smaller than its upper bound $\sum_{i=1}^{d} \|g_{1:T,i}\|_2 << dG_\infty\sqrt{T}$ and $\sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} << dG_\infty\sqrt{T}$, in particular if the class of function and data features are in the form of section 1.2 in (Duchi et al., 2011). Their results for the expected value $\mathbb{E}[\sum_{i=1}^{d} \|g_{1:T,i}\|_2]$ also apply to Adam. In particular, the adaptive method, such as Adam and Adagrad, can achieve $O(\log d\sqrt{T})$, an improvement over $O(\sqrt{dT})$ for the non-adaptive method. Decaying $\beta_{1,t}$ towards zero is important in our theoretical analysis and also matches previous empirical findings, e.g. (Sutskever et al., 2013) suggests reducing the momentum coefficient in the end of training can improve convergence.

Finally, we can show the average regret of Adam converges,

Corollary 4.2. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in R^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, ..., T\}$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$\frac{R(T)}{T} = O(\frac{1}{\sqrt{T}})$$

This result can be obtained by using Theorem 4.1 and $\sum_{i=1}^{d} \|g_{1:T,i}\|_2 \leq dG_\infty\sqrt{T}$. Thus, $\lim_{T\to\infty} \frac{R(T)}{T} = 0$.

5 RELATED WORK

Optimization methods bearing a direct relation to Adam are RMSProp (Tieleman & Hinton, 2012; Graves, 2013) and AdaGrad (Duchi et al., 2011); these relationships are discussed below. Other stochastic optimization methods include vSGD (Schaul et al., 2012), AdaDelta (Zeiler, 2012) and the natural Newton method from Roux & Fitzgibbon (2010), all setting stepsizes by estimating curvature

4 收敛性分析

我们借助(Zinkevich, 2003)提出的在线学习框架来分析Adam算法的收敛性。给定一组任意且未知的凸成本函数序列 $f_1(\theta)$，$f_2(\theta)$,...，$f_T(\theta)$，在每个时间点 t，我们的目标是预测参数 θ_t，并将其应用到之前尚未知晓的成本函数 f_t上进行评估。由于该序列的性质事先无法得知，因此我们通过“遗憾度”来衡量算法的性能，所谓遗憾度指的是所有先前在线预测值 $f_t(\theta_t)$与可行集 $\mathcal{X}$ 中所有先前步骤对应的最佳固定点参数 $f_t(\theta^*)$之间的差异之和。具体而言，遗憾度的定义如下：

$$R(T) = \sum_{t=1}^{T} [f_t(\theta_t) - f_t(\theta^*)] \tag{5}$$

其中 $\theta^* = \arg\min_{\theta \in \mathcal{X}} \sum_{t=1}^{T} f_t(\theta)$。我们证明Adam算法具有 $O(\sqrt{T})$的遗憾界，相关证明详见附录。我们的结果与针对这一通用凸型在线学习问题目前已知的最佳界限相当。此外，我们还通过一些定义来简化符号表示，此处 g_t、$\nabla f_t(\theta_t)$以及$g_{t,i}$被视作 i^{th} 元素。我们定义$g_{1:t,i} \in \mathbb{R}^t$ 为一个向量，它包含了从初始时刻到 t、$g_{1:t,i} = [g_{1,i}, g_{2,i}, \cdots, g_{t,i}]$所有迭代过程中梯度的 i^{th} 维度信息。同时，我们还定义了 γ和 $\frac{\beta_1^2}{\sqrt{\beta_2}}$。以下定理成立的条件是：学习率 α_t 以 $t^{-\frac{1}{2}}$ 的速率衰减，且一阶运行平均系数$\beta_{1,t}$ 会随着 λ呈指数级衰减，而 λ的值通常接近1，例如 $1 - 10^{-8}$。

定理 4.1 *假设该函数的梯度是有界的，即对于所有的 m、$n \in \{1, ..., T\}$和 β_1、$\beta_2 \in [0, 1)$与()都是有界的，同时由Adam算法生成的任意两点之间的距离也是有界的，即$\|\theta_m - \theta_n\|_\infty \leq D_\infty$，且满足 $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$的) 条件。设 $\alpha_t = \frac{\alpha}{\sqrt{t}}$和$\beta_{1,t} = \beta_1 \lambda^{t-1}$、$\lambda \in$ (0，1)。对于所有的 $T \geq 1$，Adam算法都能给出如下的保证。*

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Our Theorem 4.1 implies when the data features are sparse and bounded gradients, the summation term can be much smaller than its upper bound $\sum_{i=1}^{d} \|g_{1:T,i}\|_2 << dG_\infty\sqrt{T}$ and $\sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} << dG_\infty\sqrt{T}$, in particular if the class of function and data features are in the form of section 1.2 in (Duchi et al., 2011). Their results for the expected value $\mathbb{E}[\sum_{i=1}^{d} \|g_{1:T,i}\|_2]$ also apply to Adam. In particular, the adaptive method, such as Adam and Adagrad, can achieve $O(\log d\sqrt{T})$, an improvement over $O(\sqrt{dT})$ for the non-adaptive method. Decaying $\beta_{1,t}$ towards zero is important in our theoretical analysis and also matches previous empirical findings, e.g. (Sutskever et al., 2013) suggests reducing the momentum coefficient in the end of training can improve convergence.

最后，我们可以证明Adam算法的平均遗憾值是收敛的。

推论 4.2 *假设该函数的梯度是有界的，即对于所有的 m、$n \in \{1, ..., T\}$，()与()也都是有界的，同时由Adam算法生成的任意两点之间的距离也是有界的，即$\|\theta_m - \theta_n\|_\infty \leq D_\infty$。在此前提下，Adam算法能够为所有的$T \geq 1$提供如下的保证。*

$$\frac{R(T)}{T} = O(\frac{1}{\sqrt{T}})$$

这一结果可以通过运用定理4.1以及$\sum_{i=1}^{d} \|g_{1:T,i}\|_2 \leq dG_\infty\sqrt{T}$来推导得出。因此，$\lim_{T\to\infty} \frac{R(T)}{T} = 0$。

5 相关工作

与Adam直接相关的优化方法包括RMSProp（Tieleman与Hinton，2012；Graves，2013）以及AdaGrad（Duchi等人，2011）；这些关联将在下文中进行探讨。其他随机优化方法还有vSGD（Schaul等人，2012）、AdaDelta（Zeiler，2012），以及Roux与Fitzgibbon（2010）提出的自然牛顿法，这类方法均通过估算曲率来设定步长。

from first-order information. The Sum-of-Functions Optimizer (SFO) (Sohl-Dickstein et al., 2014) is a quasi-Newton method based on minibatches, but (unlike Adam) has memory requirements linear in the number of minibatch partitions of a dataset, which is often infeasible on memory-constrained systems such as a GPU. Like natural gradient descent (NGD) (Amari, 1998), Adam employs a preconditioner that adapts to the geometry of the data, since $\widehat{v}_t$ is an approximation to the diagonal of the Fisher information matrix (Pascanu & Bengio, 2013); however, Adam's preconditioner (like AdaGrad's) is more conservative in its adaption than vanilla NGD by preconditioning with the square root of the inverse of the diagonal Fisher information matrix approximation.

RMSProp: An optimization method closely related to Adam is RMSProp (Tieleman & Hinton, 2012). A version with momentum has sometimes been used (Graves, 2013). There are a few important differences between RMSProp with momentum and Adam: RMSProp with momentum generates its parameter updates using a momentum on the rescaled gradient, whereas Adam updates are directly estimated using a running average of first and second moment of the gradient. RMSProp also lacks a bias-correction term; this matters most in case of a value of β_2 close to 1 (required in case of sparse gradients), since in that case not correcting the bias leads to very large stepsizes and often divergence, as we also empirically demonstrate in section 6.4.

AdaGrad: An algorithm that works well for sparse gradients is AdaGrad (Duchi et al., 2011). Its basic version updates parameters as $\theta_{t+1} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^{t} g_t^2}$. Note that if we choose β_2 to be infinitesimally close to 1 from below, then $\lim_{\beta_2 \to 1} \widehat{v}_t = t^{-1} \cdot \sum_{i=1}^{t} g_t^2$. AdaGrad corresponds to a version of Adam with $\beta_1 = 0$, infinitesimal $(1 - \beta_2)$ and a replacement of α by an annealed version $\alpha_t = \alpha \cdot t^{-1/2}$, namely $\theta_t - \alpha \cdot t^{-1/2} \cdot \widehat{m}_t / \sqrt{\lim_{\beta_2 \to 1} \widehat{v}_t} = \theta_t - \alpha \cdot t^{-1/2} \cdot g_t / \sqrt{t^{-1} \cdot \sum_{i=1}^{t} g_t^2} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^{t} g_t^2}$. Note that this direct correspondence between Adam and Adagrad does not hold when removing the bias-correction terms; without bias correction, like in RMSProp, a β_2 infinitesimally close to 1 would lead to infinitely large bias, and infinitely large parameter updates.

6 EXPERIMENTS

To empirically evaluate the proposed method, we investigated different popular machine learning models, including logistic regression, multilayer fully connected neural networks and deep convolutional neural networks. Using large models and datasets, we demonstrate Adam can efficiently solve practical deep learning problems.

We use the same parameter initialization when comparing different optimization algorithms. The hyper-parameters, such as learning rate and momentum, are searched over a dense grid and the results are reported using the best hyper-parameter setting.

6.1 EXPERIMENT: LOGISTIC REGRESSION

We evaluate our proposed method on L2-regularized multi-class logistic regression using the MNIST dataset. Logistic regression has a well-studied convex objective, making it suitable for comparison of different optimizers without worrying about local minimum issues. The stepsize α in our logistic regression experiments is adjusted by $1/\sqrt{t}$ decay, namely $\alpha_t = \frac{\alpha}{\sqrt{t}}$ that matches with our theoratical prediction from section 4. The logistic regression classifies the class label directly on the 784 dimension image vectors. We compare Adam to accelerated SGD with Nesterov momentum and Adagrad using minibatch size of 128. According to Figure 1, we found that the Adam yields similar convergence as SGD with momentum and both converge faster than Adagrad.

As discussed in (Duchi et al., 2011), Adagrad can efficiently deal with sparse features and gradients as one of its main theoretical results whereas SGD is low at learning rare features. Adam with $1/\sqrt{t}$ decay on its stepsize should theoratically match the performance of Adagrad. We examine the sparse feature problem using IMDB movie review dataset from (Maas et al., 2011). We pre-process the IMDB movie reviews into bag-of-words (BoW) feature vectors including the first 10,000 most frequent words. The 10,000 dimension BoW feature vector for each review is highly sparse. As suggested in (Wang & Manning, 2013), 50% dropout noise can be applied to the BoW features during

基于一阶信息。函数求和优化器（SFO）（Sohl-Dickstein等人，2014年）是一种基于小批量的拟牛顿方法，但（与Adam不同）它对内存的需求与数据集的小批量划分数量成线性关系，这在GPU这类内存受限的系统中往往难以实现。与自然梯度下降法（NGD）（Amari，1998年）类似，Adam也采用了能够适应数据几何结构的预处理器，因为 $\widehat{v}_t$ 实际上是费希尔信息矩阵对角线的近似值（Pascanu & Bengio，2013年）；不过，Adam的预处理器（与AdaGrad的类似）在适配过程中更为保守，它是通过使用费希尔信息矩阵对角线近似值的逆的平方根来进行预处理的。

RMSProp：与Adam密切相关的优化方法便是RMSProp算法（Tieleman与Hinton，2012）。有时也会使用加入动量机制的版本（Graves，2013）。带有动量的RMSProp与Adam之间存在几处重要差异：带有动量的RMSProp是通过對缩放后的梯度应用动量来生成参数更新值的，而Adam则是直接利用梯度的一阶和二阶矩的滑动平均值来计算更新值。此外，RMSProp还缺乏偏差校正项；当 β_2 的值接近1时（在梯度稀疏的情况下是必需的），这一缺失就会造成极大的步长，进而引发收敛失败，我们在6.4节中也通过实验证实了这一点。

AdaGrad：一种对稀疏梯度表现良好的算法是AdaGrad（Duchi等人，2011年）。其基本版本会按照 $\theta_{t+1} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^{t} g_t^2}$的方式更新参数。需要注意的是，如果我们选择 β_2 使其略小于1且无限接近1，那么$\lim_{\beta_2 \to 1} \widehat{v}_t = t^{-1} \cdot \sum_{i=1}^{t} g_t^2$。AdaGrad实际上相当于Adam算法的一种变体，其中 $\beta_1 = 0$设为无穷小量$(1 - \beta_2)$，同时 α 被经过退火处理的版本$\alpha_t = \alpha \cdot t^{-1/2}$所替代，即 $\theta_t - \alpha \cdot t^{-1/2} \cdot \widehat{m}_t / \sqrt{\lim_{\beta_2 \to 1} \widehat{v}_t} = \theta_t - \alpha \cdot t^{-1/2} \cdot g_t / \sqrt{t^{-1} \cdot} \cdot \sum_{i=1}^{t} g_t^2 = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^{t} g_t^2}$。需注意的是，当去掉偏差校正项后，Adam与AdaGrad之间的这种直接对应关系就不再成立；在没有偏差校正的情况下，就像在RMSProp中一样，一个略小于1且无限接近1的 β_2值会导致极大的偏差以及极大的参数更新幅度。

6 实验

为了对所提出的方法进行实证评估，我们研究了多种流行的机器学习模型，包括逻辑回归、多层全连接神经网络以及深度卷积神经网络。通过使用大型模型和数据集，我们证明了Adam算法能够高效地解决实际的深度学习问题。

在比较不同的优化算法时，我们采用相同的参数初始化方式。对于学习率与动量等超参数，会在密集的网格范围内进行搜索，并以最优的超参数设置作为结果呈现。

6.1 实验：LOGISTICREGRESSION

我们在基于MNIST数据集的L2正则化多类逻辑回归问题上评估了所提出的方法。逻辑回归具有经过深入研究的凸目标函数，这一特性使其非常适合用于比较不同的优化器，且无需担心局部最小值的问题。在我们的逻辑回归实验中，步长 α 是通过 $1/\sqrt{t}$ 衰减机制来调整的，具体数值为 $\alpha_t = \frac{\alpha}{\sqrt{t}}$，该值与第4节中的理论预测相符。逻辑回归会直接对784维的图像向量进行类别标签的分类。我们采用了小批量大小为128的条件，将Adam算法与带有Nesterov动量的加速SGD以及AdaGrad算法进行了对比。如图1所示，我们发现Adam算法的收敛速度与带动量的SGD相当，且两者的收敛速度都快于AdaGrad算法。

如(Duchi等人，2011年)所述，AdaGrad算法的一个主要理论优势在于其能够高效处理稀疏特征与梯度，而SGD优化器在学习稀有特征方面的能力较弱。理论上，对步长应用$1/\sqrt{t}$ 衰减机制的Adam算法应能与AdaGrad算法的性能相当。我们利用(Maas等人，2011年)提供的IMDB电影评论数据集来研究稀疏特征问题。首先将这些电影评论预处理为词袋模型(BoW)特征向量，仅包含出现频率最高的10,000个单词。由此生成的每条评论对应的10,000维BoW特征向量都具有极高的稀疏性。正如(Wang和Manning，2013年)所建议的，可以在处理这些BoW特征时加入50%的丢弃机制噪声。

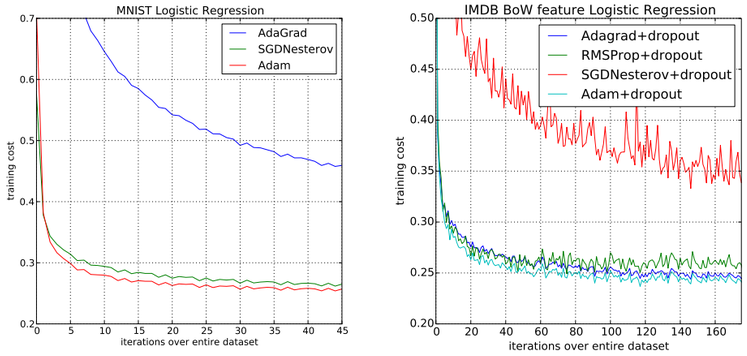


Figure 1: Logistic regression training negative log likelihood on MNIST images and IMDB movie reviews with 10,000 bag-of-words (BoW) feature vectors.

training to prevent over-fitting. In figure 1, Adagrad outperforms SGD with Nesterov momentum by a large margin both with and without dropout noise. Adam converges as fast as Adagrad. The empirical performance of Adam is consistent with our theoretical findings in sections 2 and 4. Similar to Adagrad, Adam can take advantage of sparse features and obtain faster convergence rate than normal SGD with momentum.

6.2 Experiment: Multi-layer Neural Networks

Multi-layer neural network are powerful models with non-convex objective functions. Although our convergence analysis does not apply to non-convex problems, we empirically found that Adam often outperforms other methods in such cases. In our experiments, we made model choices that are consistent with previous publications in the area; a neural network model with two fully connected hidden layers with 1000 hidden units each and ReLU activation are used for this experiment with minibatch size of 128.

First, we study different optimizers using the standard deterministic cross-entropy objective function with L_2 weight decay on the parameters to prevent over-fitting. The sum-of-functions (SFO) method (Sohl-Dickstein et al., 2014) is a recently proposed quasi-Newton method that works with minibatches of data and has shown good performance on optimization of multi-layer neural networks. We used their implementation and compared with Adam to train such models. Figure 2 shows that Adam makes faster progress in terms of both the number of iterations and wall-clock time. Due to the cost of updating curvature information, SFO is 5-10x slower per iteration compared to Adam, and has a memory requirement that is linear in the number minibatches.

Stochastic regularization methods, such as dropout, are an effective way to prevent over-fitting and often used in practice due to their simplicity. SFO assumes deterministic subfunctions, and indeed failed to converge on cost functions with stochastic regularization. We compare the effectiveness of Adam to other stochastic first order methods on multi-layer neural networks trained with dropout noise. Figure 2 shows our results; Adam shows better convergence than other methods.

6.3 Experiment: Convolutional Neural Networks

Convolutional neural networks (CNNs) with several layers of convolution, pooling and non-linear units have shown considerable success in computer vision tasks. Unlike most fully connected neural nets, weight sharing in CNNs results in vastly different gradients in different layers. A smaller learning rate for the convolution layers is often used in practice when applying SGD. We show the effectiveness of Adam in deep CNNs. Our CNN architecture has three alternating stages of 5x5 convolution filters and 3x3 max pooling with stride of 2 that are followed by a fully connected layer of 1000 rectified linear hidden units (ReLU's). The input image are pre-processed by whitening, and

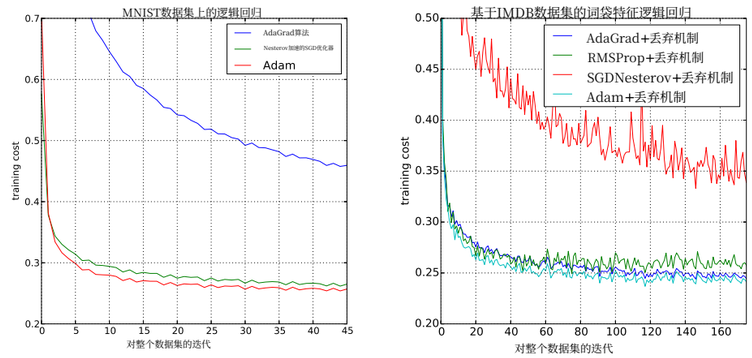


图1：在MNIST数据集和IMDB数据集上的逻辑回归训练负对数似然值。 e
使用包含10,000个词袋特征向量的评估结果。

通过训练来防止过拟合。如图1所示，无论是否加入丢弃机制带来的噪声，AdaGrad算法相比带有Nesterov动量的SGD优化器都拥有巨大的优势。Adam算法的收敛速度与AdaGrad相当。其实际表现与我们第2节和第4节中的理论研究结果一致。与AdaGrad类似，Adam算法也能利用稀疏特征，从而获得比普通带动量的SGD更快的收敛速度。

6.2 实验：多层神经网络

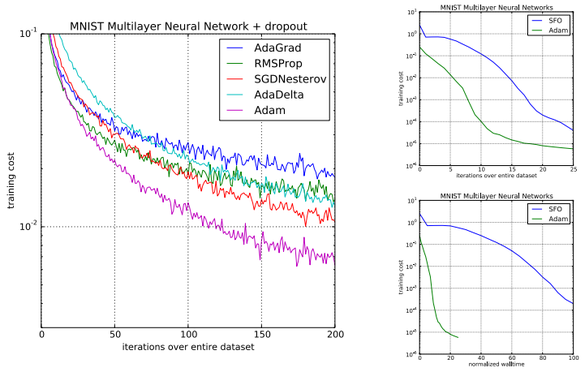
多层神经网络是具有非凸目标函数的强大模型。尽管我们的收敛性分析不适用于非凸问题，但通过实验证明，在这类情况下Adam算法往往能优于其他方法。在我们的实验中，模型选择与该领域的先前研究结果保持一致：本次实验采用了包含两个全连接隐藏层、每层拥有1000个隐藏单元并采用ReLU激活函数的神经网络模型，小批量大小设置为128。

首先，我们使用标准的确定性交叉熵目标函数，并对参数施加 L_2 权重衰减以防止过拟合，以此来研究不同的优化器。函数求和法（SFO）（Sohl-Dickstein等人，2014年）是一种近期提出的拟牛顿法，它基于数据的小批量进行运算，在多层神经网络的优化任务中表现优异。我们采用了该方法的实现版本，并将其与Adam算法进行对比以训练此类模型。图2显示，无论从迭代次数还是实际运行时间来看，Adam算法的进展都更为迅速。由于更新曲率信息需要耗费大量计算资源，SFO每轮迭代的速度比Adam慢5到10倍，而且其内存需求与小批量数量呈线性关系。

诸如丢弃机制之类的随机正则化方法是一种有效的防止过拟合的手段，因其实现简单而在实际应用中颇为常见。SFO假设子函数为确定性函数，因此在包含随机正则化的成本函数问题上无法实现收敛。我们对比了Adam算法与其他随机一阶方法在经过丢弃噪声训练的多层神经网络上的效果，图2展示了实验结果——Adam算法的收敛性能明显优于其他方法。

6.3 实验：卷积神经网络 ORKS

具备多层卷积、池化操作以及非线性单元的卷积神经网络（CNN）在计算机视觉任务中已取得了显著成效。与大多数全连接神经网络不同，CNN中的权重共享特性会导致各层产生的梯度存在巨大差异。在实际应用SGD优化器时，通常会为卷积层设置较低的学习率。本文展示了Adam算法在深度CNN中的应用效果。我们所设计的CNN架构包含三个交替阶段：首先使用5x5的卷积滤波器，接着进行步长为2的3x3最大值池化，之后再接入一个包含1000个修正线性单元（即ReLU单元）的全连接层。输入图像在处理前会先经过白化预处理。



(a)

(b)

Figure 2: Training of multilayer neural networks on MNIST images. (a) Neural networks using dropout stochastic regularization. (b) Neural networks with deterministic cost function. We compare with the sum-of-functions (SFO) optimizer (Sohl-Dickstein et al., 2014)

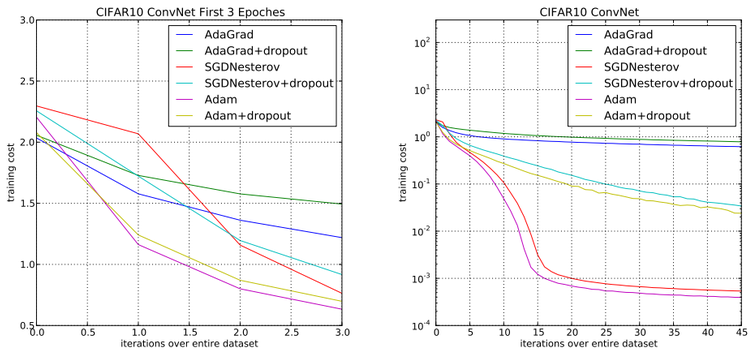
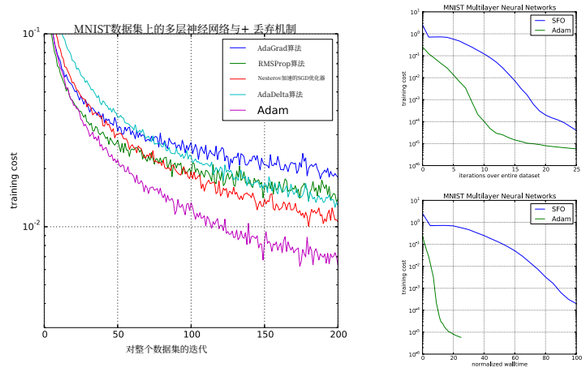


Figure 3: Convolutional neural networks training cost. (left) Training cost for the first three epochs. (right) Training cost over 45 epochs. CIFAR-10 with c64-c64-c128-1000 architecture.

dropout noise is applied to the input layer and fully connected layer. The minibatch size is also set to 128 similar to previous experiments.

Interestingly, although both Adam and Adagrad make rapid progress lowering the cost in the initial stage of the training, shown in Figure 3 (left), Adam and SGD eventually converge considerably faster than Adagrad for CNNs shown in Figure 3 (right). We notice the second moment estimate $\widehat{v}_t$ vanishes to zeros after a few epochs and is dominated by the ϵ in algorithm 1. The second moment estimate is therefore a poor approximation to the geometry of the cost function in CNNs comparing to fully connected network from Section 6.2. Whereas, reducing the minibatch variance through the first moment is more important in CNNs and contributes to the speed-up. As a result, Adagrad converges much slower than others in this particular experiment. Though Adam shows marginal improvement over SGD with momentum, it adapts learning rate scale for different layers instead of hand picking manually as in SGD.



(a)

(b)

图2：在MNIST图像上训练多层神经网络。(a)采用随机丢弃正则化的神经网络；(b)使用确定性成本函数的神经网络。我们还将这些模型与函数求和优化器（Sohl-Dickstein等人，2014年）进行了对比。

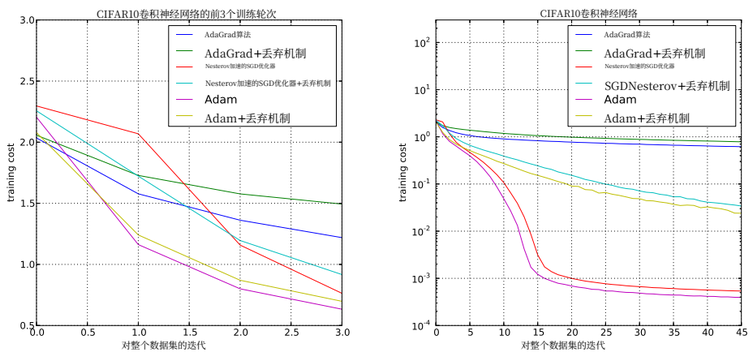
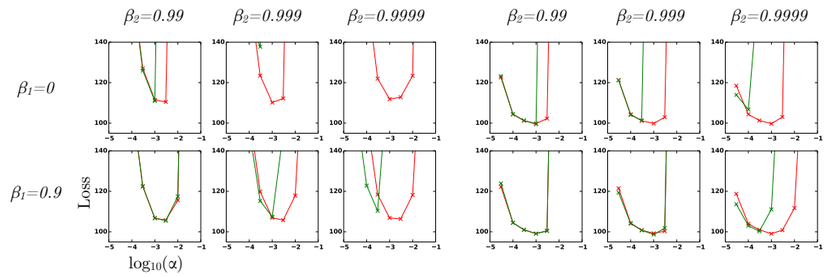


图3：卷积神经网络的训练成本。(左)前三个训练轮次的成本情况；(右)45个训练轮次的总成本。测试数据为采用c64-c64-c128-1000架构的CIFAR-10数据集。

在输入层和全连接层中均应用了丢弃机制以生成噪声。与之前的实验类似，此次研究也将小批量大小设定为128。

Interestingly, although both Adam and Adagrad make rapid progress lowering the cost in the initial stage of the training, shown in Figure 3 (left), Adam and SGD eventually converge considerably faster than Adagrad for CNNs shown in Figure 3 (right). We notice the second moment estimate $\widehat{v}_t$ vanishes to zeros after a few epochs and is dominated by the ϵ in algorithm 1. The second moment estimate is therefore a poor approximation to the geometry of the cost function in CNNs comparing to fully connected network from Section 6.2. Whereas, reducing the minibatch variance through the first moment is more important in CNNs and contributes to the speed-up. As a result, Adagrad converges much slower than others in this particular experiment. Though Adam shows marginal improvement over SGD with momentum, it adapts learning rate scale for different layers instead of hand picking manually as in SGD.



(a) after 10 epochs (b) after 100 epochs

Figure 4: Effect of bias-correction terms (red line) versus no bias correction terms (green line) after 10 epochs (left) and 100 epochs (right) on the loss (y-axes) when learning a Variational Auto-Encoder (VAE) (Kingma & Welling, 2013), for different settings of stepsize α (x-axes) and hyper-parameters β_1 and β_2.

6.4 Experiment: bias-correction term

We also empirically evaluate the effect of the bias correction terms explained in sections 2 and 3. Discussed in section 5, removal of the bias correction terms results in a version of RMSProp (Tieleman & Hinton, 2012) with momentum. We vary the β_1 and β_2 when training a variational auto-encoder (VAE) with the same architecture as in (Kingma & Welling, 2013) with a single hidden layer with 500 hidden units with softplus nonlinearities and a 50-dimensional spherical Gaussian latent variable. We iterated over a broad range of hyper-parameter choices, i.e. $\beta_1 \in [0, 0.9]$ and $\beta_2 \in [0.99, 0.999, 0.9999]$, and $\log_{10}(\alpha) \in [-5, ..., -1]$. Values of β_2 close to 1, required for robustness to sparse gradients, results in larger initialization bias; therefore we expect the bias correction term is important in such cases of slow decay, preventing an adverse effect on optimization.

In Figure 4, values β_2 close to 1 indeed lead to instabilities in training when no bias correction term was present, especially at first few epochs of the training. The best results were achieved with small values of $(1 - \beta_2)$ and bias correction; this was more apparent towards the end of optimization when gradients tends to become sparser as hidden units specialize to specific patterns. In summary, Adam performed equal or better than RMSProp, regardless of hyper-parameter setting.

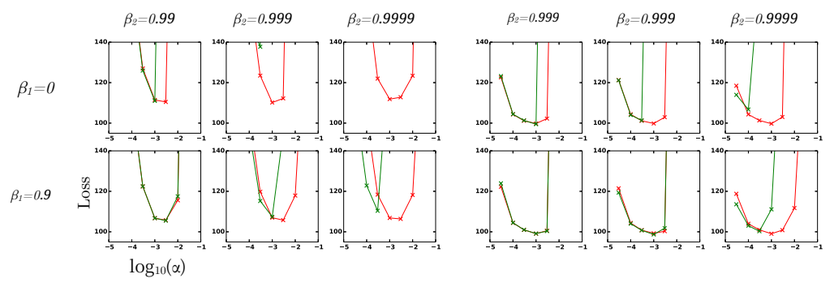
7 Extensions

7.1 AdaMax

In Adam, the update rule for individual weights is to scale their gradients inversely proportional to a (scaled) L^2 norm of their individual current and past gradients. We can generalize the L^2 norm based update rule to a L^p norm based update rule. Such variants become numerically unstable for large p. However, in the special case where we let $p \to \infty$, a surprisingly simple and stable algorithm emerges; see algorithm 2. We'll now derive the algorithm. Let, in case of the L^p norm, the stepsize at time t be inversely proportional to $v_t^{1/p}$, where:

$$v_t = \beta_2^p v_{t-1} + (1 - \beta_2^p)|g_t|^p \tag{6}$$

$$= (1 - \beta_2^p) \sum_{i=1}^{t} \beta_2^{p(t-i)} \cdot |g_i|^p \tag{7}$$



(a) 进行10个训练轮次后 (b) 进行100个训练轮次后

图4：在学习变分自编码器（VAE）（Kingma与Welling，2013）时，不同步长 α （x轴）以及超参数 β_1 和 β_2设定下，经过10个训练轮次（左图）和100个训练轮次（右图）后，偏差校正项（红线）与未应用偏差校正项（绿线）对损失值（y轴）的影响。

6.4 实验：偏差校正项

We also empirically evaluate the effect of the bias correction terms explained in sections 2 and 3. Discussed in section 5, removal of the bias correction terms results in a version of RMSProp (Tieleman & Hinton, 2012) with momentum. We vary the β_1 and β_2 when training a variational auto-encoder (VAE) with the same architecture as in (Kingma & Welling, 2013) with a single hidden layer with 500 hidden units with softplus nonlinearities and a 50-dimensional spherical Gaussian latent variable. We iterated over a broad range of hyper-parameter choices, i.e. $\beta_1 \in [0, 0.9]$ and $\beta_2 \in [0.99, 0.999, 0.9999]$, and $\log_{10}(\alpha) \in [-5, ..., -1]$. Values of β_2 close to 1, required for robustness to sparse gradients, results in larger initialization bias; therefore we expect the bias correction term is important in such cases of slow decay, preventing an adverse effect on optimization.

如图4所示，当不存在偏差校正项时，接近1的 β_2 值确实会导致训练过程中出现不稳定现象，尤其是在训练的最初几轮次。通过采用较小的$(1 - \beta_2)$值并加入偏差校正，能够获得最佳训练效果；在优化过程接近尾声时，这一优势更为明显，因为随着隐藏单元逐渐专门化以处理特定模式，梯度会变得越来越稀疏。综上所述，无论超参数如何设置，Adam算法的表现都至少与RMSProp算法相当，甚至更优。

7 扩展内容

7.1 AdaMax算法

在Adam算法中，各权重值的更新规则为：将其梯度按照与各自当前及历史梯度的（标准化后的） L^2 范数成反比的比例进行缩放。我们可以将基于 L^2 范数的更新规则推广为基于 a L^p 范数的更新规则。对于较大的p值，这类变体会在数值计算上出现不稳定现象。然而，在令 $p \to \infty$取特定值的特殊情况下，会出现一种简洁且稳定的算法——详见算法2。现在我们将推导该算法。以 L^p 范数为例，假设在时间 t 时的步长与 $v_t^{1/p}$ 成反比，其中：

$$v_t = \beta_2^p v_{t-1} + (1 - \beta_2^p)|g_t|^p \tag{6}$$

$$= (1 - \beta_2^p) \sum_{i=1}^{t} \beta_2^{p(t-i)} \cdot |g_i|^p \tag{7}$$

Algorithm 2: *AdaMax*, a variant of Adam based on the infinity norm. See section 7.1 for details. Good default settings for the tested machine learning problems are $\alpha = 0.002$, $\beta_1 = 0.9$ and $\beta_2 = 0.999$. With β_1^t we denote β_1 to the power t. Here, $(\alpha/(1-\beta_1^t))$ is the learning rate with the bias-correction term for the first moment. All operations on vectors are element-wise.

Require: α: Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0: Initial parameter vector
  $m_0 \leftarrow 0$ (Initialize 1st moment vector)
  $u_0 \leftarrow 0$ (Initialize the exponentially weighted infinity norm)
  $t \leftarrow 0$ (Initialize timestep)
  while θ_t not converged **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
    $u_t \leftarrow \max(\beta_2 \cdot u_{t-1}, |g_t|)$ (Update the exponentially weighted infinity norm)
    $\theta_t \leftarrow \theta_{t-1} - (\alpha/(1-\beta_1^t)) \cdot m_t/u_t$ (Update parameters)
  end while
  return θ_t (Resulting parameters)

Note that the decay term is here equivalently parameterised as β_2^p instead of β_2. Now let $p \to \infty$, and define $u_t = \lim_{p\to\infty}(v_t)^{1/p}$, then:

$$u_t = \lim_{p\to\infty}(v_t)^{1/p} = \lim_{p\to\infty}\left((1-\beta_2^p)\sum_{i=1}^{t}\beta_2^{p(t-i)}\cdot|g_i|^p\right)^{1/p} \tag{8}$$

$$= \lim_{p\to\infty}(1-\beta_2^p)^{1/p}\left(\sum_{i=1}^{t}\beta_2^{p(t-i)}\cdot|g_i|^p\right)^{1/p} \tag{9}$$

$$= \lim_{p\to\infty}\left(\sum_{i=1}^{t}\left(\beta_2^{(t-i)}\cdot|g_i|\right)^p\right)^{1/p} \tag{10}$$

$$= \max\left(\beta_2^{t-1}|g_1|, \beta_2^{t-2}|g_2|, \ldots, \beta_2|g_{t-1}|, |g_t|\right) \tag{11}$$

Which corresponds to the remarkably simple recursive formula:

$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|) \tag{12}$$

with initial value $u_0 = 0$. Note that, conveniently enough, we don't need to correct for initialization bias in this case. Also note that the magnitude of parameter updates has a simpler bound with AdaMax than Adam, namely: $|\Delta_t| \leq \alpha$.

7.2 Temporal averaging

Since the last iterate is noisy due to stochastic approximation, better generalization performance is often achieved by averaging. Previously in Moulines & Bach (2011), Polyak-Ruppert averaging (Polyak & Juditsky, 1992; Ruppert, 1988) has been shown to improve the convergence of standard SGD, where $\bar{\theta}_t = \frac{1}{t}\sum_{k=1}^{n}\theta_k$. Alternatively, an exponential moving average over the parameters can be used, giving higher weight to more recent parameter values. This can be trivially implemented by adding one line to the inner loop of algorithms 1 and 2: $\bar{\theta}_t \leftarrow \beta_2 \cdot \bar{\theta}_{t-1} + (1-\beta_2)\theta_t$, with $\bar{\theta}_0 = 0$. Initalization bias can again be corrected by the estimator $\widehat{\theta}_t = \bar{\theta}_t/(1-\beta_2^t)$.

8 Conclusion

We have introduced a simple and computationally efficient algorithm for gradient-based optimization of stochastic objective functions. Our method is aimed towards machine learning problems with

算法2： *AdaMax*，这是一种基于无穷范数的Adam算法变体。详情请参见第7.1节。针对所测试的机器学习问题，良好的默认参数设置为 $\alpha = 0.002$、$\beta_1 = 0.9$ 和$\beta_2 = 0.999$。其中 β_1^t 用于表示 β_1 的 t 次方。此处，$(\alpha/(1-\beta_1^t))$为包含第一时刻偏差校正项的学习率。所有向量运算均为逐元素进行。

Require: α: Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0: Initial parameter vector
  $m_0 \leftarrow 0$ (Initialize 1st moment vector)
  $u_0 \leftarrow 0$ (Initialize the exponentially weighted infinity norm)
  $t \leftarrow 0$ (Initialize timestep)
  while θ_t not converged **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
    $u_t \leftarrow \max(\beta_2 \cdot u_{t-1}, |g_t|)$ (Update the exponentially weighted infinity norm)
    $\theta_t \leftarrow \theta_{t-1} - (\alpha/(1-\beta_1^t)) \cdot m_t/u_t$ (Update parameters)
  end while
  返回 θ_t （最终参数）

请注意，此处的衰减项被等价地参数化为β_2^p 而非 β_2。现在令 $p \to \infty$，并定义 $u_t =$ 为li m$_{p\to\infty}(v_t)^{1/p}$，那么：

$$u_t = \lim_{p\to\infty}(v_t)^{1/p} = \lim_{p\to\infty}\left((1-\beta_2^p)\sum_{i=1}^{t}\beta_2^{p(t-i)}\cdot|g_i|^p\right)^{1/p} \tag{8}$$

$$= \lim_{p\to\infty}(1-\beta_2^p)^{1/p}\left(\sum_{i=1}^{t}\beta_2^{p(t-i)}\cdot|g_i|^p\right)^{1/p} \tag{9}$$

$$= \lim_{p\to\infty}\left(\sum_{i=1}^{t}\left(\beta_2^{(t-i)}\cdot|g_i|\right)^p\right)^{1/p} \tag{10}$$

$$= \max\left(\beta_2^{t-1}|g_1|, \beta_2^{t-2}|g_2|, \ldots, \beta_2|g_{t-1}|, |g_t|\right) \tag{11}$$

这对应着一个极为简洁的递归公式：

$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|) \tag{12}$$

其初始值为 $u_0 = 0$。值得注意的是，幸运的是，在这种情况下我们无需对初始化偏差进行校正。同时也要留意，与Adam算法相比，AdaMax算法下参数更新的幅度具有更简单的界限，即：$|\Delta_t| \leq \alpha$。

7.2 时间平均法

由于随机逼近会导致最后一次迭代出现噪声，因此通常通过取平均值来获得更好的泛化性能。在Moulines与Bach（2011）之前的研究中已经表明，Polyak-Ruppert平均法（Polyak与Juditsky，1992；Ruppert，1988）能够提升标准SGD的收敛速度，相关内容可见 $\bar{\theta}_t = \frac{1}{t}\sum_{k=1}^{n}\theta_k$。另一种方法是对参数使用指数移动平均，从而赋予较新的参数值更高的权重。这种方法可以通过在算法1和算法2的内循环中添加一行代码 $\bar{\theta}_t \leftarrow \beta_2 \cdot \bar{\theta}_{t-1} + (1-\beta_2)\theta_t$（其中 $\bar{\theta}_0 = 0$）即可轻松实现。初始化偏差同样可以通过估计器 $\widehat{\theta}_t = \bar{\theta}_t/(1-\beta_2^t)$进行修正。

8 结论

我们提出了一种简单且计算效率高的算法，用于基于梯度对随机目标函数进行优化。该方法专为具有大规模数据集和/或高维参数空间的机器学习问题而设计。

large datasets and/or high-dimensional parameter spaces. The method combines the advantages of two recently popular optimization methods: the ability of AdaGrad to deal with sparse gradients, and the ability of RMSProp to deal with non-stationary objectives. The method is straightforward to implement and requires little memory. The experiments confirm the analysis on the rate of convergence in convex problems. Overall, we found Adam to be robust and well-suited to a wide range of non-convex optimization problems in the field machine learning.

9 ACKNOWLEDGMENTS

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REFERENCES

Amari, Shun-Ichi. Natural gradient works efficiently in learning. *Neural computation*, 10(2):251–276, 1998.

Deng, Li, Li, Jinyu, Huang, Jui-Ting, Yao, Kaisheng, Yu, Dong, Seide, Frank, Seltzer, Michael, Zweig, Geoff, He, Xiaodong, Williams, Jason, et al. Recent advances in deep learning for speech research at microsoft. *ICASSP 2013*, 2013.

Duchi, John, Hazan, Elad, and Singer, Yoram. Adaptive subgradient methods for online learning and stochastic optimization. *The Journal of Machine Learning Research*, 12:2121–2159, 2011.

Graves, Alex. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.

Graves, Alex, Mohamed, Abdel-rahman, and Hinton, Geoffrey. Speech recognition with deep recurrent neural networks. In *Acoustics, Speech and Signal Processing (ICASSP), 2013 IEEE International Conference on*, pp. 6645–6649. IEEE, 2013.

Hinton, G.E. and Salakhutdinov, R.R. Reducing the dimensionality of data with neural networks. *Science*, 313 (5786):504–507, 2006.

Hinton, Geoffrey, Deng, Li, Yu, Dong, Dahl, George E, Mohamed, Abdel-rahman, Jaitly, Navdeep, Senior, Andrew, Vanhoucke, Vincent, Nguyen, Patrick, Sainath, Tara N, et al. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *Signal Processing Magazine, IEEE*, 29(6):82–97, 2012a.

Hinton, Geoffrey E, Srivastava, Nitish, Krizhevsky, Alex, Sutskever, Ilya, and Salakhutdinov, Ruslan R. Improving neural networks by preventing co-adaptation of feature detectors. *arXiv preprint arXiv:1207.0580*, 2012b.

Kingma, Diederik P and Welling, Max. Auto-Encoding Variational Bayes. In *The 2nd International Conference on Learning Representations (ICLR)*, 2013.

Krizhevsky, Alex, Sutskever, Ilya, and Hinton, Geoffrey E. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*, pp. 1097–1105, 2012.

Maas, Andrew L, Daly, Raymond E, Pham, Peter T, Huang, Dan, Ng, Andrew Y, and Potts, Christopher. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies-Volume 1*, pp. 142–150. Association for Computational Linguistics, 2011.

Moulines, Eric and Bach, Francis R. Non-asymptotic analysis of stochastic approximation algorithms for machine learning. In *Advances in Neural Information Processing Systems*, pp. 451–459, 2011.

Pascanu, Razvan and Bengio, Yoshua. Revisiting natural gradient for deep networks. *arXiv preprint arXiv:1301.3584*, 2013.

Polyak, Boris T and Juditsky, Anatoli B. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992.

该算法融合了两种近期颇受欢迎的优化方法的优点：AdaGrad算法处理稀疏梯度的能力，以及RMSProp算法处理非平稳目标函数的能力。它的实现十分简单，所需内存也很少。实验结果验证了关于凸问题收敛速度的分析。总体而言，我们发现Adam算法具备很强的鲁棒性，非常适合机器学习领域中各类非凸优化问题。

9 致谢

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参考文献

Amari, Shun-Ichi. 自然梯度在学习过程中具有高效性。神经计算，10(2)：251–276，1998年。

邓力、李金宇、黄翠婷、姚凯生、余东、Frank Seide、Michael Seltzer、Geoff Zweig、何晓东、Jason Williams等人。微软在语音研究领域中深度学习的最新进展。国际声学、语音与信号处理会议 *2013*，2013年。

D乌奇、[illegible]。[illegible]。《机器学习研究期刊》，12卷：2121–2159页，2011年。

亚历克斯·格雷夫斯 . 利用循环神经网络生成序列 . *arXiv*预印本 *arXiv:1308* .*0850*，2013年 .

亚历克斯·格雷夫斯、阿卜杜勒-拉赫曼·穆罕默德与杰弗里·辛顿。基于深度循环神经网络的语音识别。《声学、语音与信号处理（*ICASSP*），*2013*年电气和电子工程师协会国际会议》，第6645–6649页。电气和电子工程师协会，2013年。

H[illegible]。*Science*，313（5786）：[illegible]。

杰弗里·辛顿、邓力、余东、乔治·E·达尔、阿卜杜勒-拉赫曼·穆罕默德、纳夫迪普·贾伊特利、安德鲁·塞尼尔、文森特·范霍克、帕特里克·阮、塔拉·N·赛纳斯等。用于语音识别中声学建模的深度神经网络：四个研究团队的共同见解。电气和电子工程师协会《信号处理杂志》，29(6)：82–97，2012a。

杰弗里·E·辛顿、尼蒂什·斯里瓦斯塔瓦、亚历克斯·克里兹夫斯基、伊利亚·苏茨克维尔以及鲁斯兰·R·萨拉胡特迪诺夫。《通过防止特征检测器共同适应来提升神经网络性能》。*arXiv*预印本平台 *arXiv:12070580*，2012年。b. K

ingma, Diederik P 和 Welling, Max. 自编码变分贝叶斯。收录于 第二届国际学习表征会议（*ICLR*），2013年。

KRizhevsky, Alex，Sutskever, Ilya以及Hinton, Geoffrey E. 基于深度卷积神经网络的Imagenet分类研究。载于神经信息处理系统进展会议，第1097–1105页，2012年。

Maas, Andrew L，Daly, Raymond E，Pham, Peter T，Huang, Dan，Ng, Andrew Y 以及 Potts, Christopher . 用于情感分析的词向量学习。载于《计算语言学协会第*49*届年度会议：人类语言技术分会论文集-第*1*卷》，第142–150页。计算语言学协会，2011年。

M[illegible]。载于神经信息处理系统进展会议，第451–459页，2011年。

P[illegible]。*arXiv*预印本 *arXiv:1301.3584*，2013年。

Polyak, Boris T 和 Juditsky, Anatoli B. 通过平均法加速随机逼近。*SIAM*控制与优化杂志，30(4)：838–855，1992年。

Roux, Nicolas L and Fitzgibbon, Andrew W. A fast natural newton method. In *Proceedings of the 27th International Conference on Machine Learning (ICML-10)*, pp. 623–630, 2010.

Ruppert, David. Efficient estimations from a slowly convergent robbins-monro process. Technical report, Cornell University Operations Research and Industrial Engineering, 1988.

Schaul, Tom, Zhang, Sixin, and LeCun, Yann. No more pesky learning rates. *arXiv preprint arXiv:1206.1106*, 2012.

Sohl-Dickstein, Jascha, Poole, Ben, and Ganguli, Surya. Fast large-scale optimization by unifying stochastic gradient and quasi-newton methods. In *Proceedings of the 31st International Conference on Machine Learning (ICML-14)*, pp. 604–612, 2014.

Sutskever, Ilya, Martens, James, Dahl, George, and Hinton, Geoffrey. On the importance of initialization and momentum in deep learning. In *Proceedings of the 30th International Conference on Machine Learning (ICML-13)*, pp. 1139–1147, 2013.

Tieleman, T. and Hinton, G. Lecture 6.5 - RMSProp, COURSERA: Neural Networks for Machine Learning. Technical report, 2012.

Wang, Sida and Manning, Christopher. Fast dropout training. In *Proceedings of the 30th International Conference on Machine Learning (ICML-13)*, pp. 118–126, 2013.

Zeiler, Matthew D. Adadelta: An adaptive learning rate method. *arXiv preprint arXiv:1212.5701*, 2012.

Zinkevich, Martin. Online convex programming and generalized infinitesimal gradient ascent. 2003.

Roux, Nicolas L and Fitzgibbon, Andrew W. A fast natural newton method. In *Proceedings of the 27th International Conference on Machine Learning (ICML-10)*, pp. 623–630, 2010.

Ruppert, David. Efficient estimations from a slowly convergent robbins-monro process. Technical report, Cornell University Operations Research and Industrial Engineering, 1988.

Schaul, Tom, Zhang, Sixin, and LeCun, Yann. No more pesky learning rates. *arXiv preprint arXiv:1206.1106*, 2012.

Sohl-Dickstein, Jascha, Poole, Ben, and Ganguli, Surya. Fast large-scale optimization by unifying stochastic gradient and quasi-newton methods. In *Proceedings of the 31st International Conference on Machine Learning (ICML-14)*, pp. 604–612, 2014.

Sutskever, Ilya, Martens, James, Dahl, George, and Hinton, Geoffrey. On the importance of initialization and momentum in deep learning. In *Proceedings of the 30th International Conference on Machine Learning (ICML-13)*, pp. 1139–1147, 2013.

Tieleman, T. and Hinton, G. Lecture 6.5 - RMSProp, COURSERA: Neural Networks for Machine Learning. Technical report, 2012.

Wang, Sida and Manning, Christopher. Fast dropout training. In *Proceedings of the 30th International Conference on Machine Learning (ICML-13)*, pp. 118–126, 2013.

Zeiler, Matthew D. Adadelta: An adaptive learning rate method. *arXiv preprint arXiv:1212.5701*, 2012.

Zinkevich, Martin. Online convex programming and generalized infinitesimal gradient ascent. 2003.

10 APPENDIX

10.1 CONVERGENCE PROOF

Definition 10.1. *A function $f : R^d \to R$ is convex if for all x, $y \in R^d$, for all $\lambda \in [0, 1]$,*

$$\lambda f(x) + (1 - \lambda) f(y) \geq f(\lambda x + (1 - \lambda) y)$$

Also, notice that a convex function can be lower bounded by a hyperplane at its tangent.

Lemma 10.2. *If a function $f : R^d \to R$ is convex, then for all x, $y \in R^d$,*

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

The above lemma can be used to upper bound the regret and our proof for the main theorem is constructed by substituting the hyperplane with the Adam update rules.

The following two lemmas are used to support our main theorem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the ith element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the ith dimension of the gradients over all iterations till t, $g_{1:t,i} = [g_{1,i}, g_{2,i}, \cdots, g_{t,i}]$

Lemma 10.3. *Let $g_t = \nabla f_t(\theta_t)$ and $g_{1:t}$ be defined as above and bounded, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$. Then,*

$$\sum_{t=1}^{T} \sqrt{\frac{g_{t,i}^2}{t}} \leq 2 G_\infty \|g_{1:T,i}\|_2$$

Proof. We will prove the inequality using induction over T.

The base case for $T = 1$, we have $\sqrt{g_{1,i}^2} \leq 2 G_\infty \|g_{1,i}\|_2$.

For the inductive step,

$$\begin{aligned}
\sum_{t=1}^{T} \sqrt{\frac{g_{t,i}^2}{t}} &= \sum_{t=1}^{T-1} \sqrt{\frac{g_{t,i}^2}{t}} + \sqrt{\frac{g_{T,i}^2}{T}} \\
&\leq 2 G_\infty \|g_{1:T-1,i}\|_2 + \sqrt{\frac{g_{T,i}^2}{T}} \\
&= 2 G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_T^2} + \sqrt{\frac{g_{T,i}^2}{T}}
\end{aligned}$$

From, $\|g_{1:T,i}\|_2^2 - g_{T,i}^2 + \frac{g_{T,i}^4}{4\|g_{1:T,i}\|_2^2} \geq \|g_{1:T,i}\|_2^2 - g_{T,i}^2$, we can take square root of both side and have,

$$\begin{aligned}
\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\|g_{1:T,i}\|_2} \\
&\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\sqrt{T G_\infty^2}}
\end{aligned}$$

Rearrange the inequality and substitute the $\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2}$ term,

$$G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_T^2} + \sqrt{\frac{g_{T,i}^2}{T}} \leq 2 G_\infty \|g_{1:T,i}\|_2$$

□

10 附录

10.1 收敛性证明

定义10.1： 若对于所有的 x、 $y \in R^d$以及所有的 $\lambda \in [0, 1]$，函数 f： $R^d \to R$ 都满足特定条件，则称该函数为凸函数。

$$\lambda f(x) + (1 - \lambda) f(y) \geq f(\lambda x + (1 - \lambda) y)$$

此外，需要注意的是，凸函数在其切点处可以被一个超平面给出下界。

引理10.2： 若函数 f： $R^d \to R$ 是凸函数，那么对于所有的 x、 $y \in R^d$，都有.....

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

上述引理可用于对遗憾值设定上界，这也是我们为主定理i所给出的证明的依据。 s
通过用Adam算法的更新规则替换超平面来构建。

以下两个引理被用来支撑我们的主定理。此外，我们还使用了一些定义来简化符号表示，其中 g_t 、 $\nabla f_t(\theta_t)$ 以及$g_{t,i}$ 被视为 i^{th} 元素。我们定义$g_{1:t,i} \in \mathbb{R}^t$ 为一个向量，它包含了从初始时刻到 t、 $g_{1:t,}$ $_i = [g_{1,i}, g_{2,i}, \cdots, g_{t,i}]$所有迭代过程中梯度所具有的 i^{th} 维度的数值。

引理10.3。设 $g_t = \nabla f_t(\theta_t)$ 与 $g_{1:t}$ 如上所定义且具有有界性，即 $\|g_t\|_2 \leq G$、 $\|g_t\|_\infty \leq$、G_∞。那么，

$$\sum_{t=1}^{T} \sqrt{\frac{g_{t,i}^2}{t}} \leq 2 G_\infty \|g_{1:T,i}\|_2$$

证明。 我们将通过对 T 进行归纳法来证明该不等式。
对于 $T = 1$的基准情况，我们有 $\sqrt{g_{1,i}^2} \leq 2 G_\infty \|g_{1,\ i}\|_2$。

在归纳步骤中，

$$\begin{aligned}
\sum_{t=1}^{T} \sqrt{\frac{g_{t,i}^2}{t}} &= \sum_{t=1}^{T-1} \sqrt{\frac{g_{t,i}^2}{t}} + \sqrt{\frac{g_{T,i}^2}{T}} \\
&\leq 2 G_\infty \|g_{1:T-1,i}\|_2 + \sqrt{\frac{g_{T,i}^2}{T}} \\
&= 2 G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_T^2} + \sqrt{\frac{g_{T,i}^2}{T}}
\end{aligned}$$

由$\|g_{1:T,\ i}\|_2^2 - g_{T,\ i}^2 + \frac{g_T^4}{4\|g_{1:T}\ _{,\ i}\|_2^2} \geq \|g_{1:T}\ _{,\ i}\|_2^2 - g_{T,\ i}^2$可知，对等式两边取平方根后可得：

$$\begin{aligned}
\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\|g_{1:T,i}\|_2} \\
&\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\sqrt{T G_\infty^2}}
\end{aligned}$$

重新整理该不等式，然后代入 $\sqrt{\|g_{1:T,\ i}\|_2^2 - g_{T,\ i}^2}$ 这三个项。

$$G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_T^2} + \sqrt{\frac{g_{T,i}^2}{T}} \leq 2 G_\infty \|g_{1:T,i}\|_2$$

□

Lemma 10.4. *Let* $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. *For* $\beta_1, \beta_2 \in [0,1)$ *that satisfy* $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ *and bounded* g_t, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$, *the following inequality holds*

$$\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} \leq \frac{2}{1-\gamma} \frac{1}{\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

Proof. Under the assumption, $\frac{\sqrt{1-\beta_2^t}}{(1-\beta_1^t)^2} \leq \frac{1}{(1-\beta_1)^2}$. We can expand the last term in the summation using the update rules in Algorithm 1,

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} &= \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(\sum_{k=1}^{T}(1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T\sum_{j=1}^{T}(1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^{T} \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T\sum_{j=1}^{T}(1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^{T} \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T(1-\beta_2)\beta_2^{T-k} g_{k,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(1-\beta_1)^2}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^{T} T\left(\frac{\beta_1^2}{\sqrt{\beta_2}}\right)^{T-k} \|g_{k,i}\|_2 \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{T}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^{T} \gamma^{T-k} \|g_{k,i}\|_2
\end{aligned}$$

Similarly, we can upper bound the rest of the terms in the summation.

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} &\leq \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T-t} t\gamma^j \\
&\leq \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T} t\gamma^j
\end{aligned}$$

For $\gamma < 1$, using the upper bound on the arithmetic-geometric series, $\sum_t t\gamma^t < \frac{1}{(1-\gamma)^2}$:

$$\sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T} t\gamma^j \leq \frac{1}{(1-\gamma)^2\sqrt{1-\beta_2}} \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t}}$$

Apply Lemma 10.3,

$$\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} \leq \frac{2G_\infty}{(1-\gamma)^2\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

□

To simplify the notation, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Intuitively, our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ, that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 10.5. *Assume that the function* f_t *has bounded gradients,* $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ *for all* $\theta \in R^d$ *and distance between any* θ_t *generated by Adam is bounded,* $\|\theta_n - \theta_m\|_2 \leq D$,

引理10.4： 设 γ、 $\frac{\beta_1^2}{\sqrt{\beta_2}}$。对于满足 $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ 条件且 g_t、 $\|g_t\|_2 \leq G$、$\|g_t\|_\infty \leq G_\infty$均有界的所有 β_1、 $\beta_2 \in [0,1)$， 以下不等式成立。

$$\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} \leq \frac{2}{1-\gamma} \frac{1}{\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

证明。 在 $\frac{\sqrt{1-\beta_2^t}}{(1-\beta_1^t)^2} \leq \frac{1}{(1-\beta_1)^2}$的假设下，我们可以利用算法1中的更新规则来展开求和式中的最后一项。

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} &= \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(\sum_{k=1}^{T}(1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T\sum_{j=1}^{T}(1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^{T} \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T\sum_{j=1}^{T}(1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^{T} \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T(1-\beta_2)\beta_2^{T-k} g_{k,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(1-\beta_1)^2}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^{T} T\left(\frac{\beta_1^2}{\sqrt{\beta_2}}\right)^{T-k} \|g_{k,i}\|_2 \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{T}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^{T} \gamma^{T-k} \|g_{k,i}\|_2
\end{aligned}$$

同样地，我们也可以对求和式中其余的项给出上界。

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} &\leq \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T-t} t\gamma^j \\
&\leq \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T} t\gamma^j
\end{aligned}$$

对于 $\gamma < 1$，利用算术-几何级数的上界，有$\sum_t t\gamma^t < \frac{1}{(1-\gamma)^2}$：

$$\sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T} t\gamma^j \leq \frac{1}{(1-\gamma)^2\sqrt{1-\beta_2}} \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t}}$$

应用引理10.3，

$$\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} \leq \frac{2G_\infty}{(1-\gamma)^2\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

□

为简化符号表达，我们定义 γ、 $\frac{\beta_1^2}{\sqrt{\beta_2}}$。直观而言，当学习率 α_t 以 $t^{-\frac{1}{2}}$ 的速率衰减，且一阶运行平均系数$\beta_{1,t}$ 随 λ呈指数级衰减—通常 λ的值接近1，例如 $1 - 10^{-8}$时，以下定理便成立。

定理 10.5 f_t $\|\nabla f_t(\theta)\|_2 \leq G$ $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ $\theta \in R^d$ θ_t $\|\theta_n - \theta_m\|_2 \leq D$。假设该函数的梯度是有界的，即对于所有由*Adam*算法生成的点对，其梯度满足<span style='id:'6'> () <span style='id':'8'> () ，同时这些点对之间的距离也是有界的。

$\|\theta_m - \theta_n\|_\infty \leq D_\infty$ *for any* $m, n \in \{1, ..., T\}$, *and* β_1, $\beta_2 \in [0, 1)$ *satisfy* $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. *Let* $\alpha_t = \frac{\alpha}{\sqrt{t}}$ *and* $\beta_{1,t} = \beta_1 \lambda^{t-1}, \lambda \in (0, 1)$. *Adam achieves the following guarantee, for all* $T \geq 1$.

$\|\theta_m - \theta_n\|_\infty \leq D_\infty$ 对于任意满足 $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$条件的 m、 $n \in \{1, ..., T\}$以及 β_1、 $\beta_2 \in [0, 1)$，令和， $_t = \beta_1 \lambda^{t-1}$、 $\lambda \in$ （0， 1)。*Adam*算法能够为所有的 $T \geq 1$提供如下的保证。

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(\beta_1+1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Proof. Using Lemma 10.2, we have,

证明：利用引理10.2，我们可以得到……

$$f_t(\theta_t) - f_t(\theta^*) \leq g_t^T(\theta_t - \theta^*) = \sum_{i=1}^{d} g_{t,i}(\theta_{t,i} - \theta_{,i}^*)$$

From the update rules presented in algorithm 1,

根据算法1中给出的更新规则……

$$\begin{aligned}\theta_{t+1} &= \theta_t - \alpha_t \widehat{m}_t / \sqrt{\widehat{v}_t} \\ &= \theta_t - \frac{\alpha_t}{1-\beta_1^t}\left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_t}} m_{t-1} + \frac{(1-\beta_{1,t})}{\sqrt{\widehat{v}_t}} g_t\right)\end{aligned}$$

We focus on the ith dimension of the parameter vector $\theta_t \in R^d$. Subtract the scalar $\theta_{,i}^*$ and square both sides of the above update rule, we have,

我们重点研究参数向量 $\theta_t \in R^d$的 i^{th} 维。先减去标量 $\theta^*_{,\ i}$，再将上述更新规则的两边同时平方，即可得到……

$$(\theta_{t+1,i} - \theta_{,i}^*)^2 = (\theta_{t,i} - \theta_{,i}^*)^2 - \frac{2\alpha_t}{1-\beta_1^t}\left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} m_{t-1,i} + \left(1 - \frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} g_{t,i}\right)(\theta_{t,i} - \theta_{,i}^*)\right) + \alpha_t^2 \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}}\right)^2$$

We can rearrange the above equation and use Young's inequality, $ab \leq a^2/2 + b^2/2$. Also, it can be shown that $\sqrt{\widehat{v}_{t,i}} = \sqrt{\sum_{j=1}^{t}(1-\beta_2)\beta_2^{t-j} g_{j,i}^2}/\sqrt{1-\beta_2^t} \leq \|g_{1:t,i}\|_2$ and $\beta_{1,t} \leq \beta_1$. Then

我们可以重新整理上述方程并运用杨氏不等式 $ab \leq a^2/2 + b^2/2$。此外，还可以证明$\sqrt{\widehat{v}_{t,\ i}} = \sqrt{\sum_{j=1}^{t}(1-\beta_2)\beta_2^{t-j} g_{j,\ i}^2}/\sqrt{1-\beta_2^t} \leq \|g_{1:t,\ i}\|_2$，以及$\beta_{1,\ t} \leq \beta_1$。随后……

$$\begin{aligned}g_{t,i}(\theta_{t,i} - \theta_{,i}^*) =& \frac{(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2\alpha_t(1-\beta_{1,t})}\left((\theta_{t,i} - \theta_{,t}^*)^2 - (\theta_{t+1,i} - \theta_{,i}^*)^2\right) \\ &+ \frac{\beta_{1,t}}{(1-\beta_{1,t})} \frac{\widehat{v}_{t-1,i}^{\frac{1}{4}}}{\sqrt{\alpha_{t-1}}}(\theta_{,i}^* - \theta_{t,i})\sqrt{\alpha_{t-1}} \frac{m_{t-1,i}}{\widehat{v}_{t-1,i}^{\frac{1}{4}}} + \frac{\alpha_t(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2(1-\beta_{1,t})}\left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}}\right)^2 \\ \leq& \frac{1}{2\alpha_t(1-\beta_1)}\left((\theta_{t,i} - \theta_{,t}^*)^2 - (\theta_{t+1,i} - \theta_{,i}^*)^2\right)\sqrt{\widehat{v}_{t,i}} + \frac{\beta_{1,t}}{2\alpha_{t-1}(1-\beta_{1,t})}(\theta_{,i}^* - \theta_{t,i})^2\sqrt{\widehat{v}_{t-1,i}} \\ &+ \frac{\beta_1\alpha_{t-1}}{2(1-\beta_1)} \frac{m_{t-1,i}^2}{\sqrt{\widehat{v}_{t-1,i}}} + \frac{\alpha_t}{2(1-\beta_1)} \frac{\widehat{m}_{t,i}^2}{\sqrt{\widehat{v}_{t,i}}}\end{aligned}$$

We apply Lemma 10.4 to the above inequality and derive the regret bound by summing across all the dimensions for $i \in 1, ..., d$ in the upper bound of $f_t(\theta_t) - f_t(\theta^*)$ and the sequence of convex functions for $t \in 1, ..., T$:

我们将引理10.4应用于上述不等式，通过对 $f_t(\theta_t) - f_t(\theta^*)$的上界以及 $t \in 1$、... T这一系列凸函数在所有维度上求和，从而推导出遗憾界。

$$\begin{aligned}R(T) \leq& \sum_{i=1}^{d} \frac{1}{2\alpha_1(1-\beta_1)}(\theta_{1,i} - \theta_{,i}^*)^2\sqrt{\widehat{v}_{1,i}} + \sum_{i=1}^{d}\sum_{t=2}^{T} \frac{1}{2(1-\beta_1)}(\theta_{t,i} - \theta_{,i}^*)^2\left(\frac{\sqrt{\widehat{v}_{t,i}}}{\alpha_t} - \frac{\sqrt{\widehat{v}_{t-1,i}}}{\alpha_{t-1}}\right) \\ &+ \frac{\beta_1\alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \frac{\alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 \\ &+ \sum_{i=1}^{d}\sum_{t=1}^{T} \frac{\beta_{1,t}}{2\alpha_t(1-\beta_{1,t})}(\theta_{,i}^* - \theta_{t,i})^2\sqrt{\widehat{v}_{t,i}}\end{aligned}$$

From the assumption, $\|\theta_t - \theta^*\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$, we have:

$$\begin{aligned}
R(T) \leq & \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \frac{D_\infty^2}{2\alpha} \sum_{i=1}^{d} \sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t\widehat{v}_{t,i}} \\
\leq & \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 \\
& + \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha} \sum_{i=1}^{d} \sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t}
\end{aligned}$$

We can use arithmetic geometric series upper bound for the last term:

$$\begin{aligned}
\sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t} & \leq \sum_{t=1}^{t} \frac{1}{(1-\beta_1)} \lambda^{t-1} \sqrt{t} \\
& \leq \sum_{t=1}^{t} \frac{1}{(1-\beta_1)} \lambda^{t-1} t \\
& \leq \frac{1}{(1-\beta_1)(1-\lambda)^2}
\end{aligned}$$

Therefore, we have the following regret bound:

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha\beta_1(1-\lambda)^2}$$

□

基于 $\|\theta_t - \theta^*\|_2 \leq D$、 $\|\theta_m - \theta_n\|_\infty \leq D_\infty$这两个假设，我们可以得到：

$$\begin{aligned}
R(T) \leq & \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \frac{D_\infty^2}{2\alpha} \sum_{i=1}^{d} \sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t\widehat{v}_{t,i}} \\
\leq & \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 \\
& + \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha} \sum_{i=1}^{d} \sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t}
\end{aligned}$$

对于最后一项，我们可以运用算术几何级数上界。

$$\begin{aligned}
\sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t} & \leq \sum_{t=1}^{t} \frac{1}{(1-\beta_1)} \lambda^{t-1} \sqrt{t} \\
& \leq \sum_{t=1}^{t} \frac{1}{(1-\beta_1)} \lambda^{t-1} t \\
& \leq \frac{1}{(1-\beta_1)(1-\lambda)^2}
\end{aligned}$$

因此，我们可以得到如下的遗憾界。

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha\beta_1(1-\lambda)^2}$$

□